

CSS Custom Fonts

Master Font Loading, @font-face, and Typography Control

What are CSS Custom Fonts?

CSS Custom Fonts allow you to use any typeface on your website, not just system-installed fonts. You can load fonts from external sources (Google Fonts, Adobe Fonts, self-hosted), define custom font files, and have complete control over typography. This revolutionizes web design by enabling unlimited font choices.

Key Advantages

- **Unlimited Choices:** Use any font, not just web-safe fonts
- **Brand Consistency:** Use exact brand fonts across the web
- **Better Control:** Adjust weights, styles, and sizes precisely
- **Improved Design:** Custom fonts enhance visual hierarchy and appeal
- **Performance:** Load only the fonts and weights you need

Font Sources

1. **Google Fonts:** Free, reliable, easy integration
2. **Adobe Fonts:** Premium fonts with Creative Cloud subscription
3. **Self-Hosted:** Upload your own font files for complete control
4. **Font Services:** Typekit, FontAwesome, custom APIs

@font-face Rule

The @font-face CSS rule defines a custom font by specifying the font name and the source file location. This is the foundation of custom fonts on the web.

Basic @font-face Syntax

```
@font-face {
  font-family: 'MyCustomFont';
  src: url('font-file.woff2') format('woff2'),
       url('font-file.woff') format('woff');
}

body {
  font-family: 'MyCustomFont', sans-serif;
}
```

@font-face Properties

```
@font-face {
  font-family: 'CustomFont';
  src: url('font.woff2') format('woff2');
  font-weight: 400;
  font-style: normal;
  font-stretch: normal;
  font-display: swap;
  unicode-range: U+0000-00FF;
}
```

Font File Formats

Format	Extension	Browser Support	Size	Best For
WOFF2	.woff2	Modern browsers (95%+)	Smallest	Primary choice
WOFF	.woff	All modern browsers	Small	Fallback
TTF	.ttf	Most browsers	Large	Legacy support
OTF	.otf	Most browsers	Large	Professional fonts
EOT	.eot	IE only	Large	IE 6-8

💡 **Best Practice:** Use WOFF2 with WOFF fallback. This covers 99%+ of browsers while minimizing file size.

Google Fonts (Easiest Method)

Google Fonts provides a free library of 1500+ fonts with easy integration. No self-hosting required, optimized delivery, and automatic updates.

Method 1: Link Tag (Recommended)

```
<head>
<link href="https://fonts.googleapis.com/css2?family=Roboto:wght@400;700&display=swap"
      rel="stylesheet">
</head>

body { font-family: 'Roboto', sans-serif; }
```

Demo - Roboto Font:

Regular (400): The quick brown fox

Bold (700): The quick brown fox

Method 2: @import (CSS Only)

```
@import url('https://fonts.googleapis.com/css2?family=Open+Sans:wght@400;600;700&display=swap');
```

URL Parameters Explained

- family=:** Font names (use + for spaces, separate with |)
- wght=:** Weights to load (400, 600, 700, etc.)
- display=:** Font display strategy (swap, fallback, optional)
- subset=:** Character sets (latin, cyrillic, greek, etc.)

Demo - Playfair Display (Serif):

Elegant Typography

Google Fonts URL Builder

Visit fonts.google.com to browse fonts and generate custom URLs. Select fonts, weights, and styles, then copy the link code.

font-display Property

Controls how the browser displays text while a custom font is loading. Critical for performance and user experience.

font-display Values

```
@font-face {
  font-family: 'CustomFont';
  src: url('font.woff2') format('woff2');
  font-display: auto | block | swap | fallback | optional;
}
```

Value	Block Period	Swap Period	Best For
auto	3s	Infinite	Browser default
block	3s	Infinite	Important fonts (headings)
swap	0s	Infinite	Body text (fast fallback)
fallback	0.1s	3s	Optional fonts
optional	0.1s	0s	Non-critical fonts

font-display Explained

- **auto:** Let browser decide (safe, default)
- **block:** Hide text until font loads, then show (FOUT - Flash of Unstyled Text delayed)
- **swap:** Show fallback immediately, replace when font loads (FOIT - Flash of Invisible Text eliminated)
- **fallback:** Quick fallback, limited swap period (3s)
- **optional:** Quick fallback, don't wait for font load (best performance)

🔴 **Best Practice:** Use `font-display: swap` for body text (shows immediately), and `font-display: block` for important headings.

Font Weights & Styles

Different fonts support different weights and styles. Always specify which ones you need to minimize file size.

Font Weight Values

100 - Thin
200 - Extra Light
300 - Light
400 - Normal (default)
500 - Medium
600 - Semi Bold
700 - Bold
800 - Extra Bold
900 - Black

Demo - Different Font Weights:

Light 300

Normal 400

Semibold 600

Bold 700

Black 900

Font Styles

```
font-style: normal | italic | oblique;
```

Demo - Font Styles:

Normal Style

Italic Style

Oblique Style

Best Practices for Font Loading

- **Load only weights you use:** Each weight = separate font file
- **Combine weights:** Load 400 and 700 for body and bold text
- **Minimize variants:** Don't load italic if not used
- **Use variable fonts:** One file with all weights (when available)

Typography Control Properties

Line Height

Controls vertical space between lines. Critical for readability, especially for body text.

```
line-height: 1 | 1.5 | 2 | 24px | 150%;
```

Demo - Line Height Impact:

line-height: 1 - Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

line-height: 1.5 - Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

line-height: 2 - Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Letter Spacing

Controls horizontal space between characters. Useful for headlines, all-caps text, and visual hierarchy.

```
letter-spacing: -.02em | 0 | 0.05em | 2px;
```

Demo - Letter Spacing:

TIGHT

NORMAL

WIDE

WIDER

Font Size

```
font-size: 12px | 1em | 1.5em | 100%;
```

Demo - Font Sizes:

Small (0.9em)

Normal (1em)

Large (1.2em)

XLarge (1.5em)

Font Variant

Controls text variations like small caps. Adds visual distinction without changing markup.

```
font-variant: normal | small-caps | all-small-caps; font-variant-caps: normal | small-caps | all-small-caps | petite-caps | all-petite-caps; font-variant-numeric: normal | lining-nums | oldstyle-nums | tabular-nums;
```

Demo - Font Variants:

NORMAL TEXT

SMALL CAPS TEXT

Font Stacks (Fallbacks)

Always provide fallback fonts in case the custom font fails to load. CSS applies the first available font in the list.

Building a Font Stack

```
font-family: 'Playfair Display', Georgia, 'Times New Roman', serif; /* Custom → Backup serif → Serif system font → Generic serif */
```

Common Font Stacks

```
/* Professional Serif */
font-family: 'Merriweather', Georgia, serif;

/* Modern Sans-Serif */
font-family: 'Inter', -apple-system, BlinkMacSystemFont, 'Segoe UI', sans-serif;

/* Monospace */
font-family: 'Roboto Mono', 'Courier New', monospace;
```

Demo - Font Stacks:

Serif Stack

Sans-Serif Stack

Monospace Stack

Performance & Optimization

File Size Considerations

- **WOFF2:** ~60% smaller than TTF
- **Subset fonts:** Load only needed characters (latin, cyrillic, etc.)
- **Load only needed weights:** 400 + 700 is often sufficient
- **Preload critical fonts:** Use `<link rel="preload">`

Font Preloading

```
<link rel="preload" as="font" href="font.woff2" type="font/woff2" crossorigin>
```

Self-Hosting vs CDN

Aspect	Self-Hosted	Google Fonts	Adobe Fonts
Cost	Free	Free	Subscription
Control	Full	Limited	Limited
Speed	Fast (same server)	Very fast (CDN)	Fast (CDN)
Ease	Requires setup	Very easy	Easy
Selection	Any font	1500+ fonts	7000+ fonts

💡 **Recommendation:** Start with Google Fonts for simplicity. Self-host only if you need specific fonts or want maximum control.

Best Practices & Tips

🔴 **Tip 1: Use font-display: swap**

Avoids invisible text while fonts load. Content shows immediately with fallback, updates when font arrives.

🔴 **Tip 2: Limit Font Variations**

Load only weights you use (typically 400 and 700). Each variation = additional HTTP request and file size.

🔴 **Tip 3: Provide Proper Fallbacks**

Always include generic families (serif, sans-serif, monospace) in your font stack.

🔴 **Tip 4: Preload Critical Fonts**

Use `<link rel="preload">` for fonts used above the fold.

🔴 **Tip 5: Test Font Loading**

Use Chrome DevTools Network tab to see font loading times and potential FOIT/FOUT.

🔴 **Tip 6: Consider System Fonts**

Use system font stacks for faster load times: `-apple-system, BlinkMacSystemFont, 'Segoe UI'`

Browser Support

✅ **Excellent Support for Custom Fonts**

@font-face: IE 6+, all modern browsers
WOFF2: Chrome 37+, Firefox 39+, Safari 14+, Edge 15+
font-display: Chrome 60+, Firefox 58+, Safari 11+

Note: Always provide WOFF2 and WOFF formats for broad compatibility.

Common Mistakes to Avoid

❌ **Mistake 1:** Not using font-display, causing invisible text (FOIT)

✅ **Fix:** Always set `font-display: swap` to show fallback immediately.

❌ **Mistake 2:** Loading too many font weights

✅ **Fix:** Load only weights you use. Each weight is a separate file.

❌ **Mistake 3:** No fallback fonts specified

✅ **Fix:** Always end font-family with a generic family: `serif, sans-serif, monospace`

❌ **Mistake 4:** Using incorrect file paths in @font-face

✅ **Fix:** Use relative paths carefully; absolute URLs work from any CSS file location.

❌ **Mistake 5:** Not subsetting fonts for international sites

✅ **Fix:** Use unicode-range or subset parameters to load only needed characters.

Complete @font-face Reference

@font-face Properties

font-family - Name for the font
src - File path(s)
font-weight - 100-900
font-style - normal | italic
font-stretch - Width variant

font-display Values

auto - Browser default
block - Wait for font
swap - Show fallback first
fallback - Quick timeout
optional - No swap period

File Format Priorities

WOFF2 - Primary (smallest)
WOFF - Fallback
TTF - Legacy support
EOT - IE 6-8 only